

Station can
transmit.

a. *1*-Persistent

Station
can transmit.

b.
Nonpersistent

Channel
busy?

[true]

[false]

Generate a random
number ($R = 0$ to
1)

Channel
busy?

[false]

[false]

RSP

Wait
a slot

[true]

[true]

Station
can transmit.

Use backoff
process
as though collision
occurred.

c.
p-Persistent

2. With probability $q=1-p$, the station waits for the beginning of the next time slot

and checks the line again.

a. If the line is idle, it goes to step 1.

b. If the line is busy, it acts as though a collision has occurred and uses the back-off procedure.

12.1.3 CSMA/CD

The CSMA method does *not* specify the procedure following a collision. Carrier sense multiple access with collision detection (CSMA/CD) augments the algorithm to handle the collision.

In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station is finished. If, however, there is a collision, the frame is sent again.

To better understand CSMA/CD, let us look at the first bits transmitted by the two stations involved in the collision. Although each station continues to send bits in the frame until it detects the collision, we show what happens as the first bits collide. In

Figure 12.11, stations A and C are involved in the collision.

(MAC)

Figure 12.11 Collision of the first bits in CSMA/CD



At **time** f_1 , station A **has** executed its persistence procedure and starts sending the bits of its **frame**. At time f_2 , station C **has** not yet sensed **the** first bit sent by A. Station C executes its persistence procedure and starts sending the bits in its frame, which propagate both to the left and to the **right**. The collision occurs **some-** time after time f_2 . Station C **detects** a collision at **time 3** **when** it

receives the first bit of A's frame. Station C immediately (**or after** a short time, but we **assume** immediately) aborts transmission. Station **A detects** collision at time 14 when it receives the first bit of C's frame; it also **immediately aborts transmission**. Looking at the figure, we see that **A transmits** for the duration 14-11; **C transmits** for the duration 13-12.

Now that we **know** the time durations, for **the two transmissions**, we can show a more complete graph in Figure 12.12.

Figure 12.12 *Collision and **abortion** in CSMA/CD*



Minimum Frame Size

For CSMA/CD to work, **we** need **a** restriction on the frame **size**. Before sending the last bit of the **frame**, the sending station must detect a collision, if any, and abort the **transmis- sion**. This is so because **the** station, **once the entire frame is** sent, does not **keep** a copy of

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12.1.4 CSMA/CA

Carrier sense multiple access with collision avoidance (CSMA/CA) was invented for wireless networks. Collisions are avoided through **the** use of CSMA/CA's three strategies: the interframe **space**, the contention window, and acknowledgments, **a** shown in Figure 12.15. **We** discuss RTS and CTS frames **later**.

Figure 12.15 Flow *diagram of* CSMA/CA

Legend

K: Number of attempts

Tp: Backoff
time

IFS: Interframe
Space

RTS: Request to
send

CTS: Clear to **send**

$K=0$

Station **has**
a frame to send

Channel **free**?

[**false**]

[**true**]

Carrier sense

Wait IFS

RTS signal at Brint
moder/hub
Sa transmission stop
30781431- Ford 4/2
signal VATSITY
VOTE 13

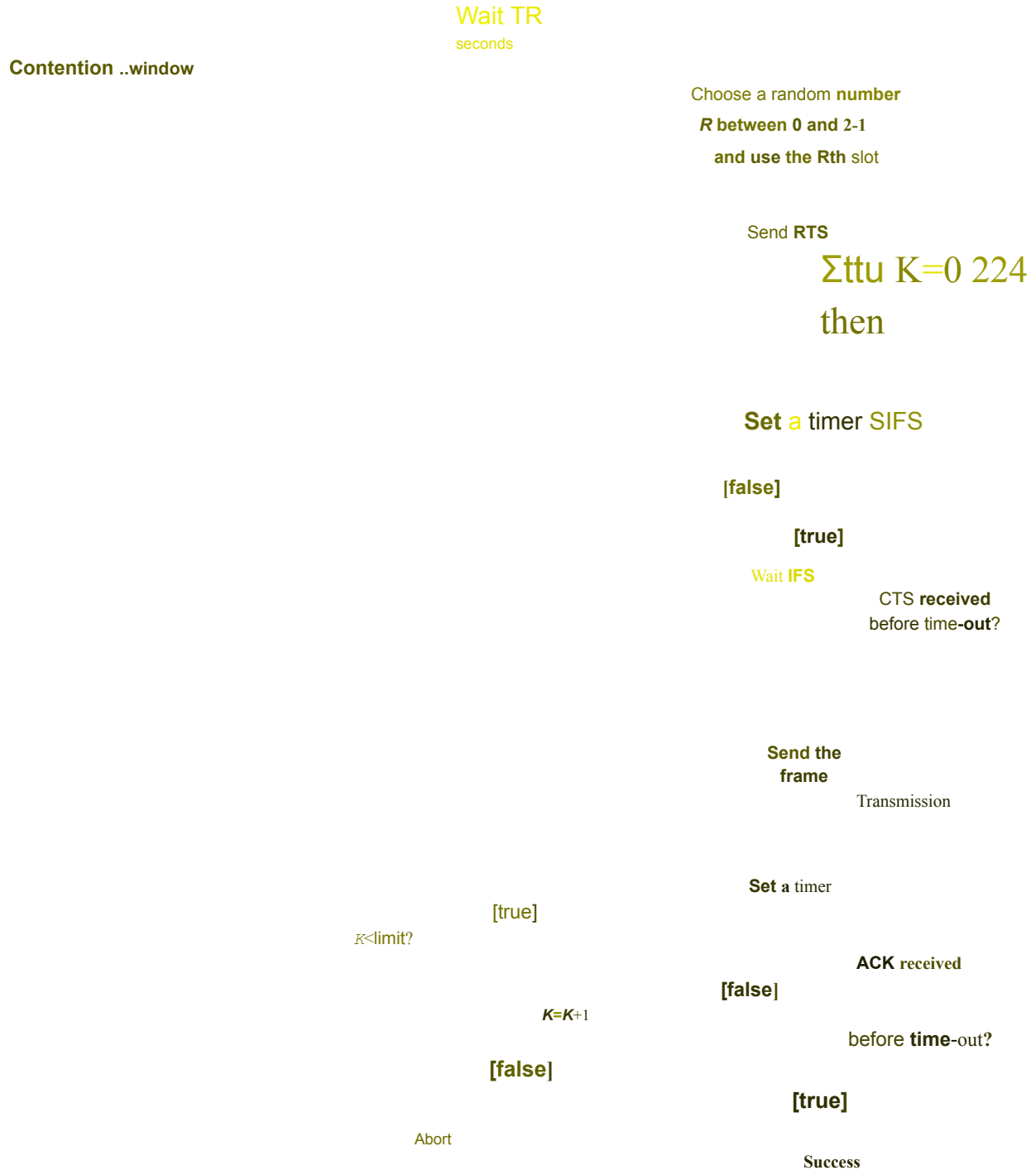
এই এই আসার আগেও 2itu

700 data Dronz

data-পাঠা

ই [237] Car? IFS

time wait



Interframe **Space (IFS)**. First, collisions **are** avoided by deferring transmission even **if** the channel is found idle, **When**

an idle channel is found, the station does not send immediately. It waits for a period of time called the *interframe space* or *IFS*. Even though the channel may appear idle when it is sensed, a distant station may have already started transmitting. The distant station's signal has not yet reached this

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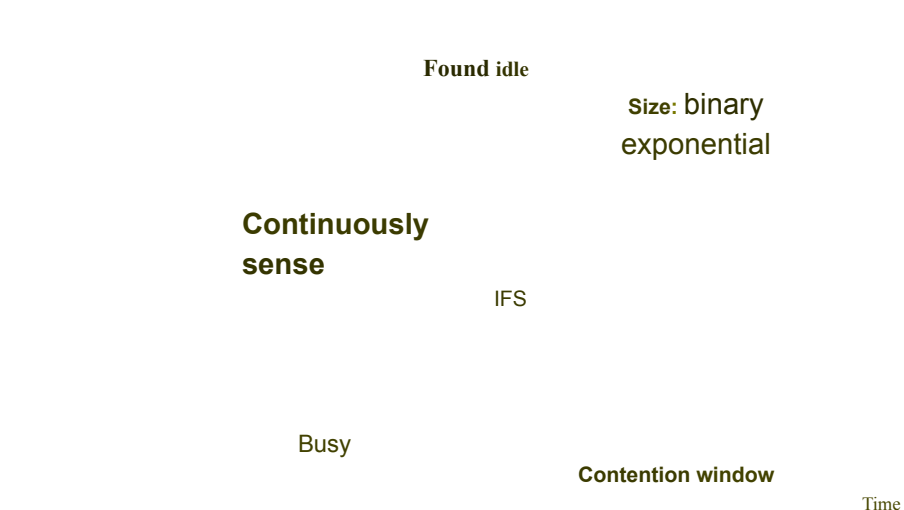
station. The IFS time allows the front of the transmitted signal by the distant station to reach this station. After waiting an IFS time, if the channel is still idle, the station can send, but it still needs to wait a time equal to the contention window (described next). The IFS variable can also be used to prioritize stations or frame types. For example, a station that is assigned a shorter IFS has a higher priority.

Contention Window. The contention window is an amount of time divided into slots. A station that is ready to send chooses a random number of slots as its wait time. The number of slots in the window changes according to the binary exponential backoff strategy. This means that it is set to one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time. **This** is similar to the p -persistent method except that a random outcome defines the very number of slots taken by the waiting station. One interesting point about the contention window is that the station needs to sense the channel after each time slot. **However**, if the station finds the channel busy, it does not restart the process; it just stops the timer and

restarts it when the channel is sensed as **idle**. This gives priority to the station with the **longest waiting** time. See Figure 12.16.

Figure 12.16 *Contention*

window



Acknowledgment. With all these precautions, there still may be a collision resulting in destroyed data. In addition, the data may be corrupted during the transmission. The positive **acknowledgment** and the time-out timer can help guarantee that the receiver has received the frame.

Frame *Exchange Time*

Line

Figure 12.17 shows the exchange of data and control frames in time.

1. Before sending a frame, the source station senses the medium by checking the energy level at the carrier frequency.
 - a. The channel uses a persistence strategy with backoff until the

channel is idle. **b.** After the station is found to be idle, the station waits for a period of time called the *DCF interframe space (DIFS)*; then the station sends a control frame called the *request to send (RTS)*.

2. After receiving the RTS and waiting a period of time called the *short interframe space (SIFS)*, the destination station sends a control frame, called the *clear to send (CTS)*, to the source station. This control frame indicates that the destination station is ready to receive data.